

Chapter 6

Hydrogen Atom

SCHRÖDINGER'S EQUATION FOR THE HYDROGEN ATOM

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m}{\hbar^2} (E - U) \psi = 0$$

$$\text{Electric Potential energy } U = -\frac{e^2}{4\pi\epsilon_0 r}.$$

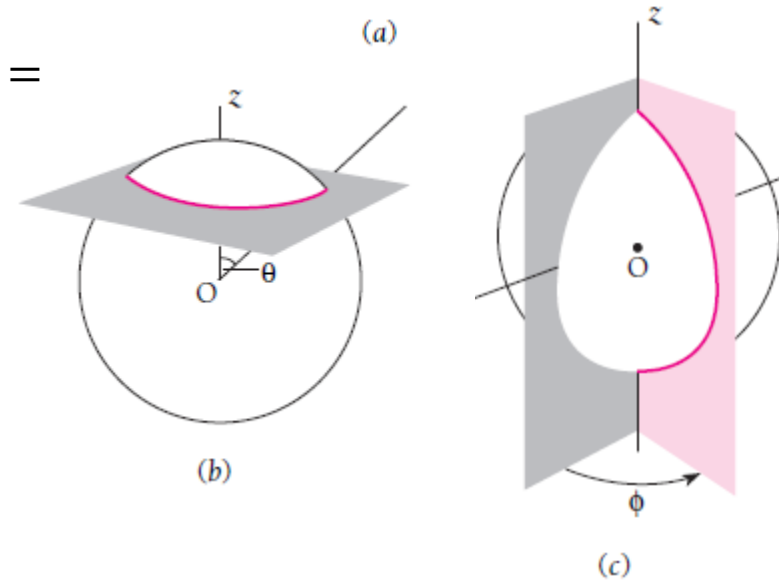
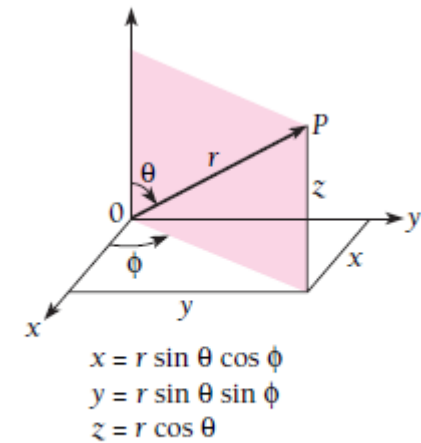
Negative sign $\rightarrow$ the potential is attractive

Spherical polar coordinates

r = length of the radius vector from origin O to point $P = \sqrt{x^2 + y^2 + z^2}$

θ = angle between the radius vector and the $+z$ axis
= zenith angle

$$= \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right) = \cos^{-1} \left(\frac{z}{r} \right)$$



Figures:

a) Spherical polar coordinates.

(b) polar angle $\rightarrow \theta \rightarrow \theta = 0$ means true north, and $\theta = \pi$ means true south.

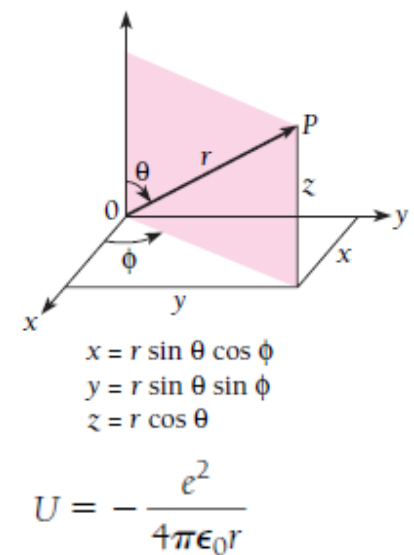
(c) Azimuth angle $\rightarrow \phi \rightarrow$ the east or west angle from the x-axis.

ϕ = angle between the projection of the radius vector in the xy plane and the $+x$ axis, measured in the direction shown

$$= \text{azimuthal angle} = \tan^{-1} \frac{y}{x}$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m}{\hbar^2} (E - U) \psi = 0$$

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$



<https://planetmath.org/DerivationOfTheLaplacianFromRectangularToSphericalCoordinates>

In spherical polar coordinates Schrödinger's equation is written

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{2m_e}{\hbar^2} (E - U) \psi = 0$$

Substitute the potential energy U and multiply the entire equation by $r^2 \sin^2 \theta$

Hydrogen Atom

$$\sin^2 \theta \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{\partial^2 \psi}{\partial \phi^2} + \frac{2m_e r^2 \sin^2 \theta}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) \psi = 0$$

Solving the polar Schrödinger equation gives three quantum numbers.
 $Q\# \rightarrow$ describe the electron in a hydrogen atom, in place of the single quantum number of the Bohr theory.

Hydrogen Atom wave function

$$\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi)$$

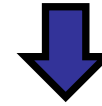
$$\psi = R\Theta\Phi$$

$$\frac{\partial \psi}{\partial r} = \Theta\Phi \frac{\partial R}{\partial r} = \Theta\Phi \frac{dR}{dr}$$

$$\frac{\partial \psi}{\partial \theta} = R\Phi \frac{\partial \Theta}{\partial \theta} = R\Phi \frac{d\Theta}{d\theta}$$

$$\frac{\partial^2 \psi}{\partial \phi^2} = R\Theta \frac{\partial^2 \Phi}{\partial \phi^2} = R\Theta \frac{d^2 \Phi}{d\phi^2}$$

solutions are **separable** into the product of three functions



it transforms a difficult 3D problem into three much easier 1D ones.

$$\frac{\sin^2 \theta}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} + \frac{2m_e r^2 \sin^2 \theta}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) = 0$$

The third term is a function of azimuth angle ϕ only, whereas the other terms are functions of r and θ only.

Rearrange

$$\frac{\sin^2 \theta}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{2m_e r^2 \sin^2 \theta}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) = -\frac{1}{\Phi} \frac{d^2 \Phi}{d\varnothing^2}$$

This eqn can be correct only if both sides of it are equal to the same constant, since they are functions of different variables .

It is convenient to call this constant m_l^2

The differential equation for the function $\varnothing$ is therefore

$$-\frac{1}{\Phi} \frac{d^2 \Phi}{d\varnothing^2} = m_l^2$$

Substitute m_l^2 for the RHS of topmost eqn on the slide, **divide the entire eqn by $\sin^2 \theta$** and rearrange the various terms

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2m_e r^2}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) = \frac{m_l^2}{\sin^2 \theta} - \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right)$$

The eqn has different variable appear on each side requiring both sides be equal to the same constant. This constant is called $l(l + 1)$. The equations for the functions Θ and R are :

$$\frac{m_l^2}{\sin^2 \theta} - \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = l(l + 1)$$

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2m_e r^2}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) = l(l + 1)$$

Equation for Φ

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} + m_l^2 \Phi = 0$$

Equation for Θ

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \left[l(l + 1) - \frac{m_l^2}{\sin^2 \theta} \right] \Theta = 0$$

Equation for R

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[\frac{2m_e r^2}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) - \frac{l(l+1)}{r^2} \right] R = 0$$

Each of these is an ordinary differential equation for a single function of a single variable.

Only the equation for R depends on the potential energy $U(r)$.

Additional NOTE

The R(r) equation can be written in two equivalent ways

Either as an equation for R(r), or for the reduced radial function P(r)=rR(r)

Define the reduced radial function: $P_{n\ell}(r) \equiv rR_{n\ell}(r)$

$$R_{n\ell}(r): \frac{1}{r^2} \cdot \frac{d}{dr} \left(r^2 \frac{dR_{n\ell}(r)}{dr} \right) - \frac{\ell(\ell+1)R_{n\ell}(r)}{r^2} - \left(\frac{2m_e}{\hbar^2} \right) [U(r) - E_n] R_{n\ell}(r) = 0$$

OR

$$\text{For } P_{n\ell}(r): \frac{d^2 P_{n\ell}}{dr^2} - \left(\frac{2m_e}{\hbar^2} \right) \left[U(r) + \frac{\ell(\ell+1)\hbar^2}{2m_e r^2} - E_n \right] P_{n\ell}(r) = 0$$

$$R_{n\ell}(r) = \frac{P_{n\ell}(r)}{r}$$

Using P(r) removes the first-derivative term and often makes the radial equation look like a 1D Schrödinger equation with an effective potential.

$R_{n\ell}(r)$: **Associated Laguerre functions.**

Additional NOTE

$$P_{n\ell}(r): \frac{d^2 P_{n\ell}}{dr^2} - \left(\frac{2m_e}{\hbar^2} \right) \left[U(r) + \frac{\ell(\ell + 1)\hbar^2}{2m_e r^2} - E_n \right] P_{n\ell}(r) = 0$$

The first term,

$$U(r) = -\frac{e^2}{4\pi\epsilon_0 r},$$

is the ordinary attractive Coulomb potential. It pulls the electron toward the nucleus.

The second term,

$$\frac{\ell(\ell + 1)\hbar^2}{2m_e r^2}$$

is called the **centrifugal potential** or **centrifugal barrier**.

It appears because the electron has angular momentum.

Classically, for a particle moving in a central potential,

$$E = \frac{p_r^2}{2m} + \frac{L^2}{2mr^2} + V(r).$$

The term

$$\frac{L^2}{2mr^2} \rightarrow \text{rotational kinetic energy.}$$

Quantum mechanically,

$$L^2 = l(l + 1)\hbar^2,$$

so

$$\frac{L^2}{2m_e r^2} \rightarrow \frac{\hbar^2 l(l + 1)}{2m_e r^2}.$$

Description of solution

- We will not go through the mathematics of the solution. We can only accept solutions for which the wave function is normalizable (does not blow up).
- Some radial solutions blow up at $r = 0$ or $r = \infty \rightarrow$ Hence discarded.
- The radial solutions $R(r) \rightarrow$ **Associated Laguerre functions**. $\rightarrow$ An exponential function $e^{-\alpha r}$, multiplied by a polynomial in r .
- The polar angular solutions $\Theta(\theta) \rightarrow$ **Legendre polynomial** $\rightarrow$ are polynomials containing powers of $\sin \theta$ and $\cos \theta$.
- The azimuthal solutions $\Phi(\phi) \rightarrow$ Periodic (have the same value at $\Phi(\phi)$ and $\Phi(\phi+2\pi)$, etc. They turn out to depend on $e^{im_l\phi}$, where m_l is an integer, either positive, negative or zero.

$$m_l = 0, \pm 1, \pm 2, \pm 3, \dots \dots \dots, \pm l$$

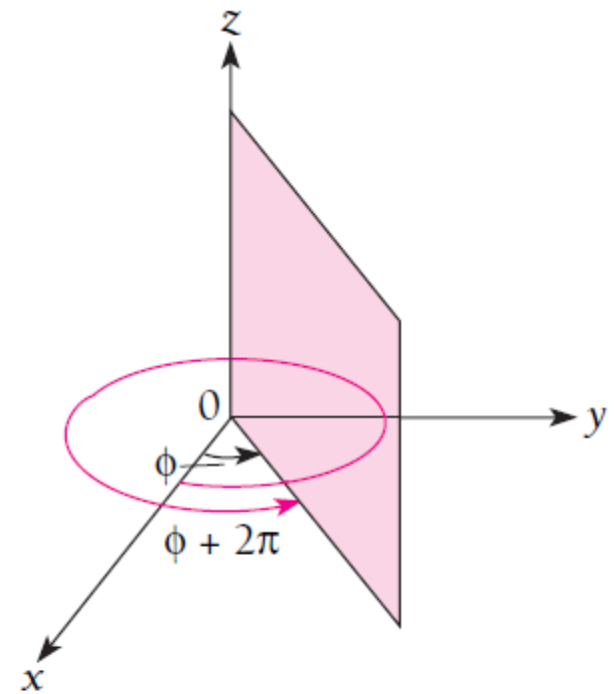


Figure 6.2 The angles ϕ and $\phi + 2\pi$ both identify the same meridian plane.

The solution for the radial part $R(r)$ of the hydrogen atom wave function Ψ also requires that a certain condition be fulfilled. This condition is that one of the negative values E_n (signifying that the electron is bound to the atom) specified by

$$E_n = -\frac{1}{(4\pi\epsilon_0)^2} \frac{m_e e^4}{2n^2 \hbar^2} = \frac{E_1}{n^2} = -\frac{13.6 \text{ eV}}{n^2}$$

$$n = 1, 2, 3 \dots \dots \dots$$

The same formula for the energy levels of the hydrogen atom that Bohr obtained.

Another condition that must be obeyed in order to solve the radial eqn is that n k/a the **Principal Q#**

$$n \geq (l + 1)$$

This requirement may be expressed as a condition on l in the form

$$l = 0, 1, 2, 3, \dots \dots \dots (n - 1)$$

Principal quantum number

$$n = 1, 2, 3, \dots$$

Orbital quantum number

$$l = 0, 1, 2, \dots, (n - 1)$$

Magnetic quantum number

$$m_l = 0, \pm 1, \pm 2, \dots, \pm l$$

n	l	m_l	$\Phi(\phi)$	$\Theta(\theta)$	$R(r)$	$\psi(r, \theta, \phi)$
1	0	0	$\frac{1}{\sqrt{2\pi}}$	$\frac{1}{\sqrt{2}}$	$\frac{2}{a_0^{3/2}} e^{-r/a_0}$	$\frac{1}{\sqrt{\pi} a_0^{3/2}} e^{-r/a_0}$
2	0	0	$\frac{1}{\sqrt{2\pi}}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2\sqrt{2} a_0^{3/2}} \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$	$\frac{1}{4\sqrt{2\pi} a_0^{3/2}} \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$
2	1	0	$\frac{1}{\sqrt{2\pi}}$	$\frac{\sqrt{6}}{2} \cos \theta$	$\frac{1}{2\sqrt{6} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$	$\frac{1}{4\sqrt{2\pi} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0} \cos \theta$
2	1	± 1	$\frac{1}{\sqrt{2\pi}} e^{\pm i\phi}$	$\frac{\sqrt{3}}{2} \sin \theta$	$\frac{1}{2\sqrt{6} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$	$\frac{1}{8\sqrt{\pi} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0} \sin \theta e^{\pm i\phi}$

$$a_0 = 4\pi\epsilon_0\hbar^2/me^2 = 5.292 \times 10^{-11} \text{ m}$$

PRINCIPAL QUANTUM NUMBER

Quantization of energy → n is quantized

ORBITAL QUANTUM NUMBER

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[\frac{2m_e}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) - \frac{l(l+1)}{r^2} \right] R = 0 \quad U = -\frac{e^2}{4\pi\epsilon_0 r}$$

$$E = KE_{radial} + KE_{orbital} + U = KE_{radial} + KE_{orbital} - \frac{e^2}{4\pi\epsilon_0 r}$$

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2m_e}{\hbar^2} \left[KE_{radial} + KE_{orbital} - \frac{\hbar^2 l(l+1)}{2m_e r^2} \right] R = 0$$

If the last two terms in the square brackets of the equation cancel each other out, we have a differential equation for $R(r)$ that involves functions of the radius vector r exclusively.

$$KE_{orbital} = \frac{\hbar^2 l(l+1)}{2m_e r^2}$$

Since the orbital kinetic energy of the electron and the magnitude of its angular momentum are

$$KE_{orbital} = \frac{1}{2} m_e v_{orbital}^2 \qquad L = m_e v_{orbital} r$$

The orbital kinetic energy

$$KE_{orbital} = \frac{L^2}{2m_e r^2}$$

Compare above with $KE_{orbital} = \frac{\hbar^2 l(l+1)}{2m_e r^2}$

$$\frac{L^2}{2m_e r^2} = \frac{\hbar^2 l(l+1)}{2m_e r^2}$$

Electron angular momentum $L = \hbar \sqrt{l(l+1)}$

$$l = 0, 1, 2, 3, \dots \dots \dots (n-1)$$

Like total energy E , *angular momentum is both conserved and quantized*.

Designation of Angular-Momentum States

It is customary to specify electron angular-momentum states by a letter, with s corresponding to $l = 0$, p to $l = 1$, and so on, according to the following scheme:

Angular-
momentum states

$l = 0$ 1 2 3 4 5 6 $\dots$
 s p d f g h i $\dots$

	$l = 0$	$l = 1$	$l = 2$	$l = 3$	$l = 4$	$l = 5$
$n = 1$	1s					
$n = 2$	2s	2p				
$n = 3$	3s	3p	3d			
$n = 4$	4s	4p	4d	4f		
$n = 5$	5s	5p	5d	5f	5g	
$n = 6$	6s	6p	6d	6f	6g	6h

MAGNETIC QUANTUM NUMBER

- The orbital quantum number l determines the magnitude L of the electron's angular momentum.
- Angular momentum is a vector quantity, meaning both its magnitude and direction must be specified.
- The vector L is perpendicular to the plane of rotational motion, following the right-hand rule.
- For a hydrogen atom, the direction in space becomes significant because an electron revolving around the nucleus acts as a minute current loop, producing a magnetic field like a magnetic dipole.
- An electron with angular momentum interacts with an external magnetic field B .
- The magnetic quantum number m_l specifies the direction of L by determining its component along the direction of the magnetic field.
- This phenomenon is called space quantization.

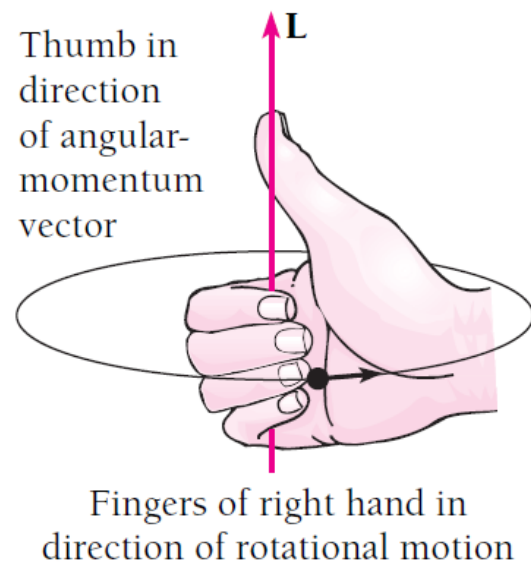


Figure 6.3 The right-hand rule for angular momentum.

If the magnetic-field direction be parallel to the z axis, the component of $\mathbf{L}$ in z direction is

Space quantization $L_z = m_l \hbar \quad m_l = 0, \pm 1, \pm 2, \dots, \pm l$

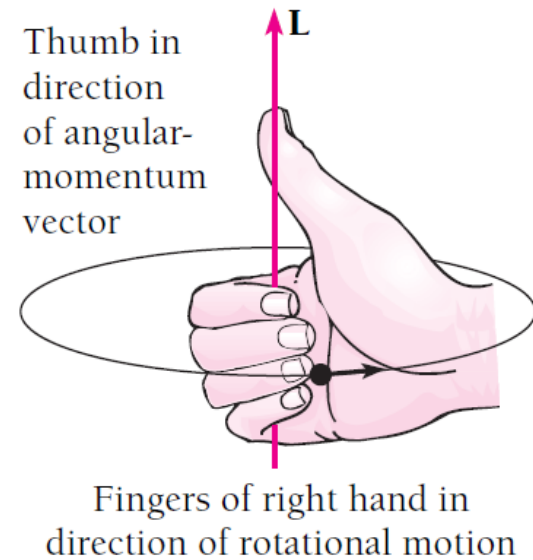
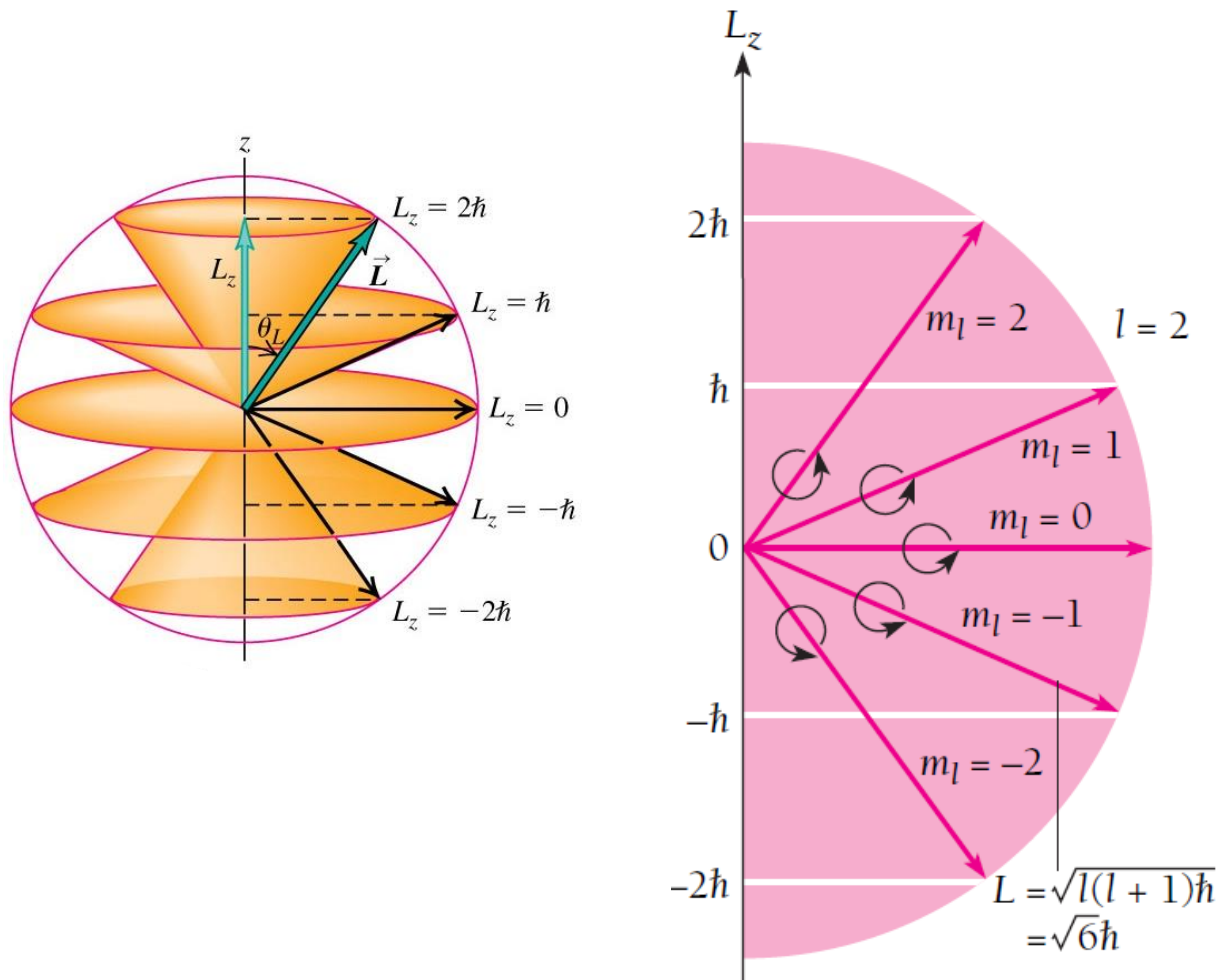


Figure 6.3 The right-hand rule for angular momentum.

ELECTRON PROBABILITY DENSITY

The probability density $|\Psi|^2$ that corresponds to the electron wave function $\Psi = R\Theta\Phi$ in the hydrogen atom is

$$|\Psi|^2 = |R|^2 |\Theta|^2 |\Phi|^2$$

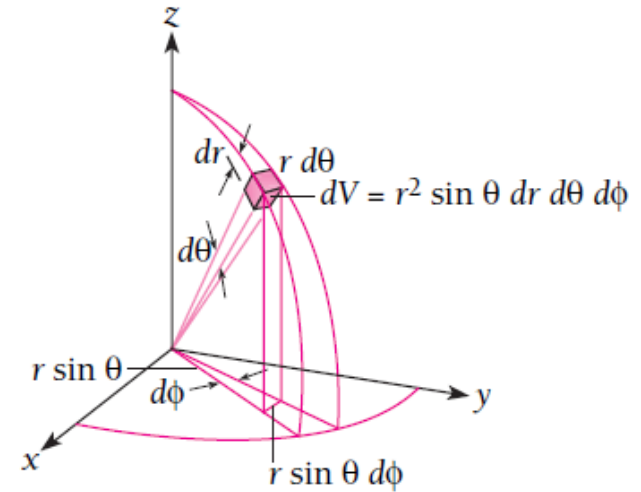
Probability of Finding the Electron

$$dP = \Psi^*(r, \theta, \phi) \Psi(r, \theta, \phi) dV$$

The differential volume element in spherical polar coordinates is

$$dV = r^2 \sin\theta dr d\theta d\phi$$

$$P(r)dr = r^2 R^*(r)R(r)dr \int_0^\pi |\Theta|^2 \sin\theta d\theta \int_0^{2\pi} |\Phi|^2 d\phi = r^2 |R|^2 dr$$



$$P(r)dr = r^2 R^*(r)R(r)dr \int_0^\pi |\Theta|^2 \sin\theta d\theta \int_0^{2\pi} |\Phi|^2 d\phi = r^2 |R|^2 dr$$

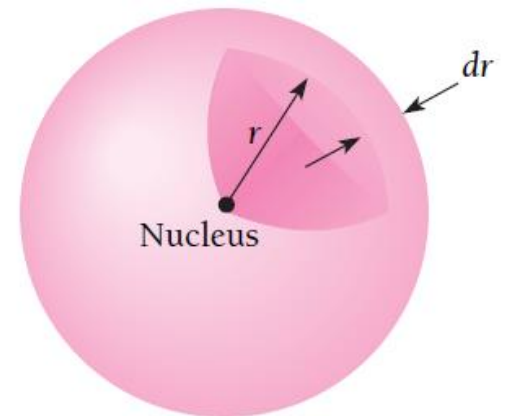
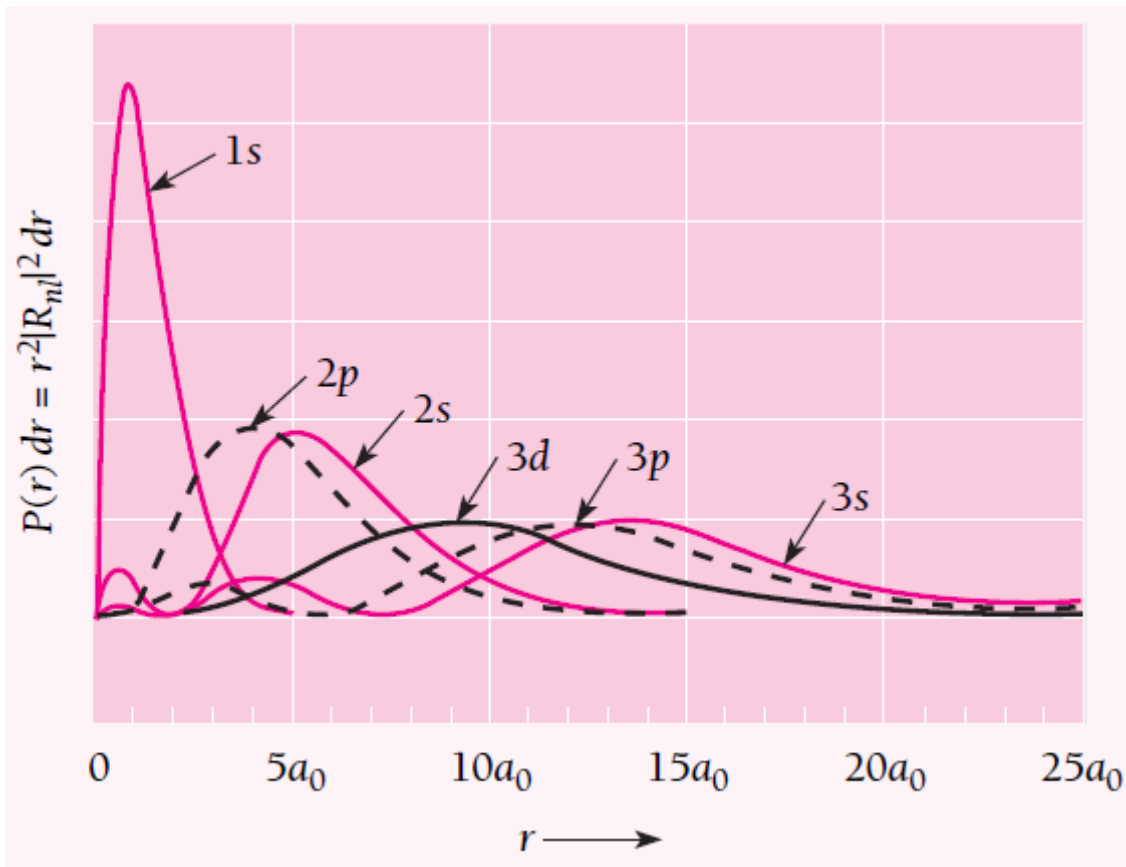
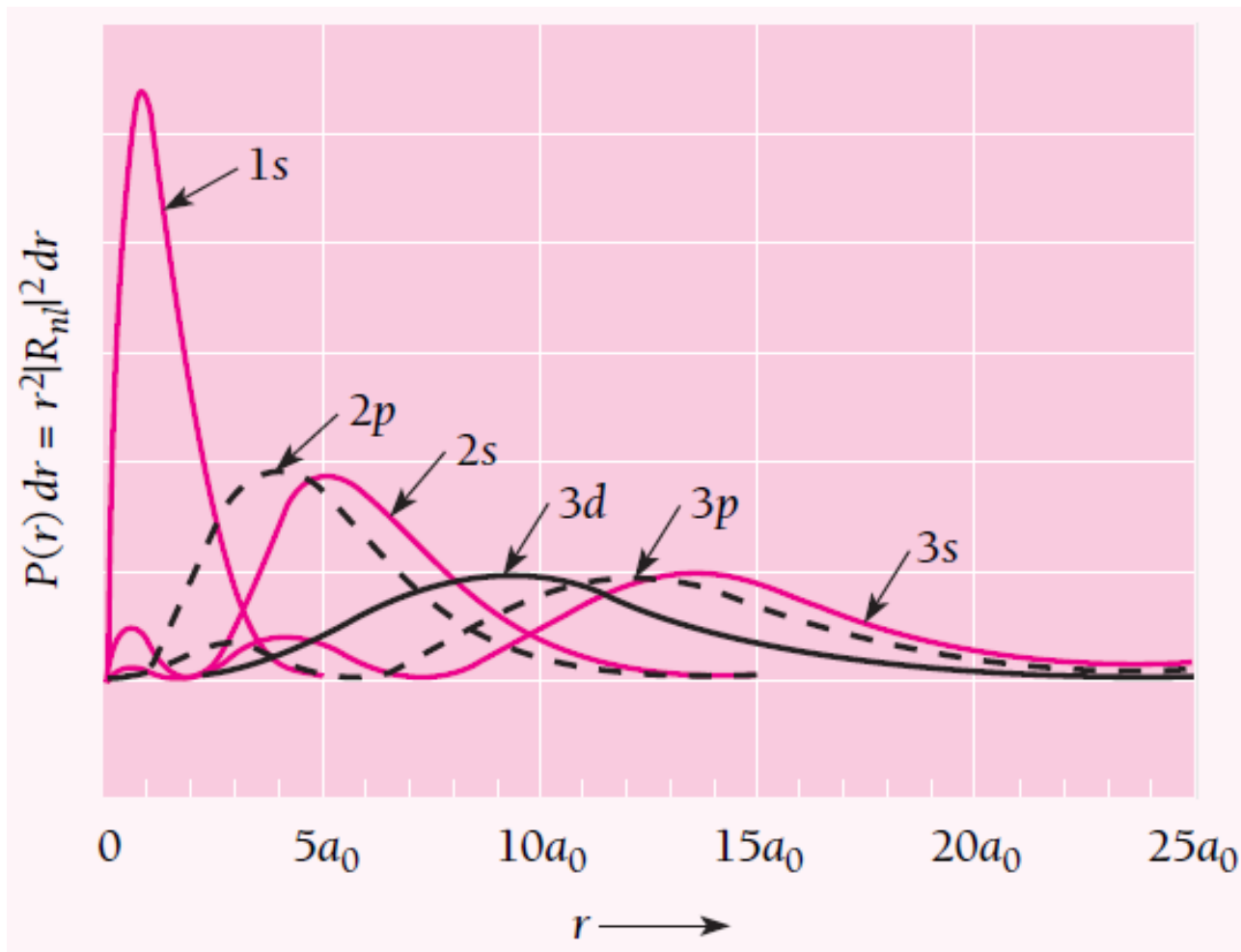


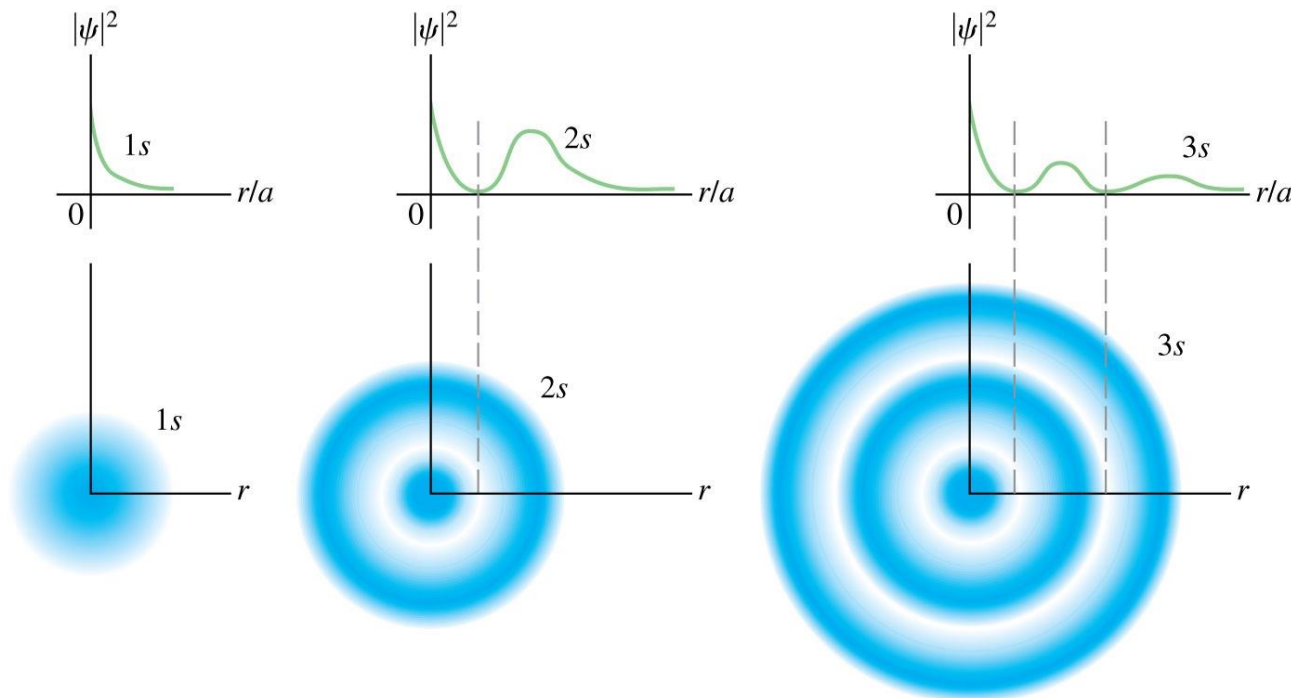
Figure 6.10 The probability of finding the electron in a hydrogen atom in the spherical shell between r and $r + dr$ from the nucleus is $P(r) dr$.

- The **most probable** value of r for a $1s$ electron is a_0 , the orbital radius of a ground-state electron in the Bohr model.
- The **average** value of r for a $1s$ electron is $1.5a_0$.



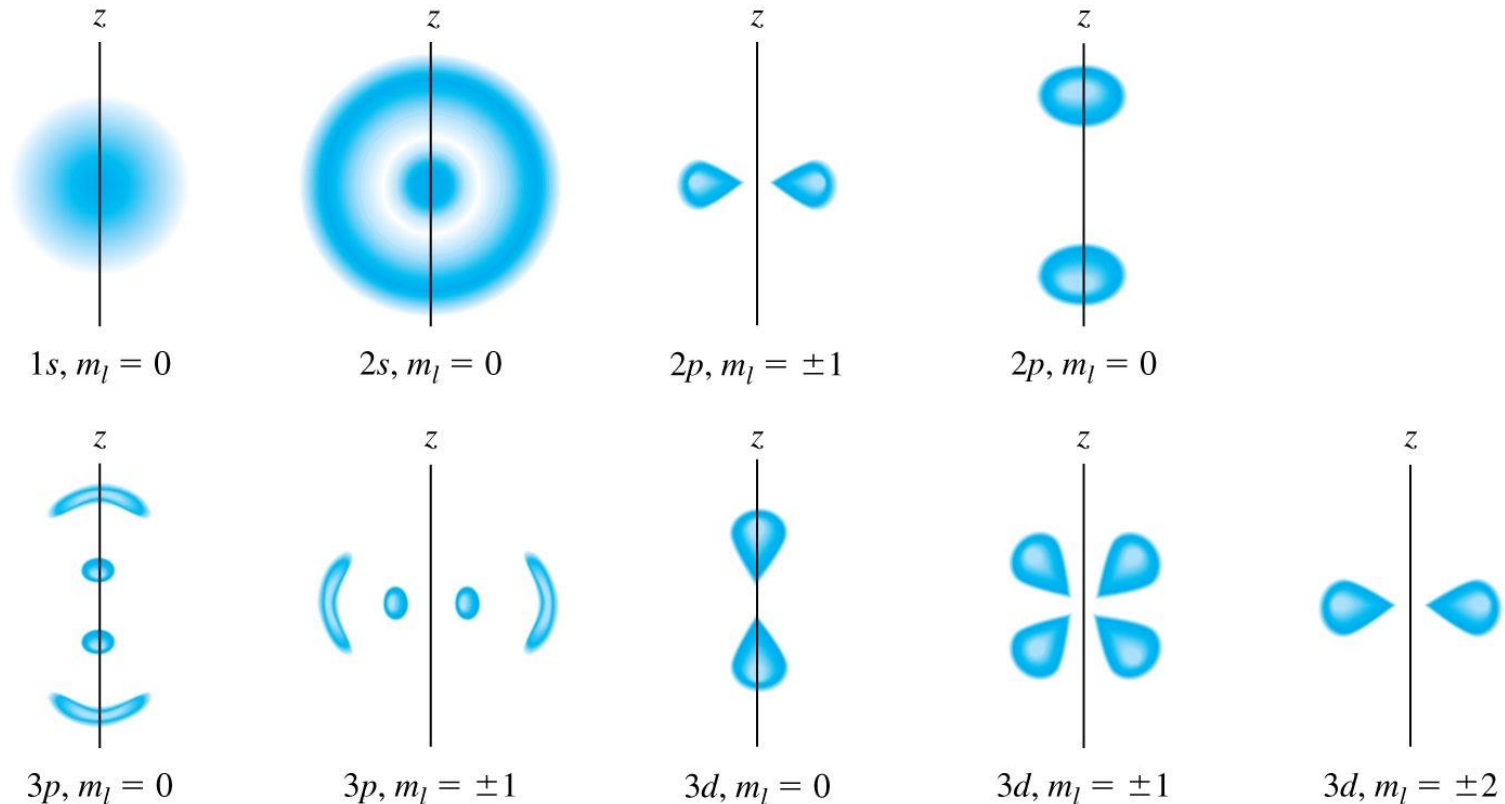
The hydrogen atom: Probability distributions I

- States of the hydrogen atom with $l = 0$ (zero orbital angular momentum) have spherically symmetric wave functions that depend on r but not on θ or ϕ . These are called s states.
- Figure below shows the electron probability distributions for three of these states.



The hydrogen atom: Probability distributions II

- States of the hydrogen atom with nonzero orbital angular momentum, such as p states ($l = 1$) and d states ($l = 2$), have wave functions that are *not* spherically symmetric.
- Figure below shows the electron probability distributions for several of these states, as well as for two spherically symmetric s states.



Number of nodes

- The total number of nodes in a hydrogen orbital is $n - 1$.
- The number of angular nodes is ℓ .
- The number of radial nodes is $n - \ell - 1$.
- Therefore, total nodes = $(n - \ell - 1) + \ell = n - 1$.

Examples

For **1s**: $n = 1, \ell = 0$

Radial nodes: $1 - 0 - 1 = 0$

Angular nodes: 0

For **2s**: $n = 2, \ell = 0$

Radial nodes: $2 - 0 - 1 = 1$

Angular nodes: 0.

For **2p**: $n = 2, \ell = 1$

Radial nodes: $2 - 1 - 1 = 0$

Angular nodes: 1

For **3d**: $n = 3, \ell = 2$

radial nodes: $3 - 2 - 1 = 0$

Angular nodes: 2

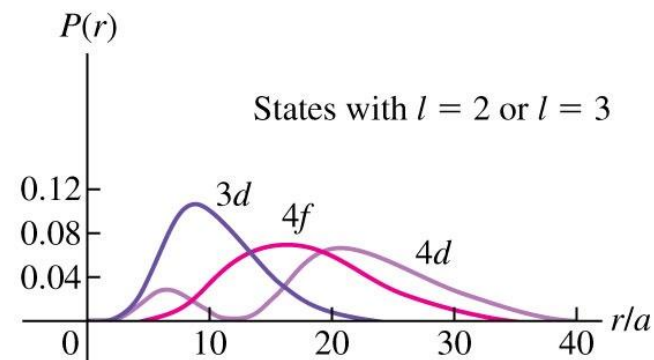
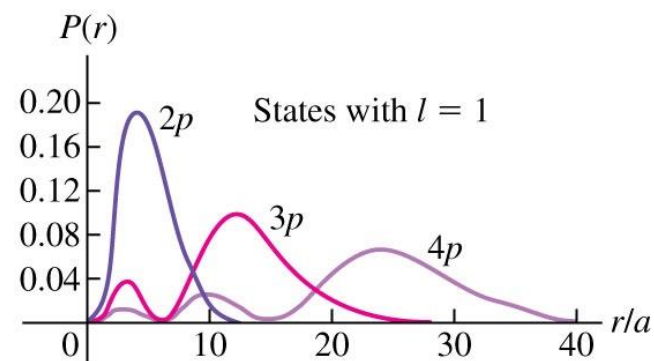
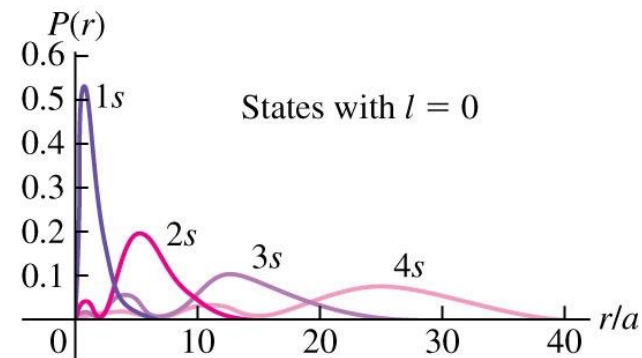
The hydrogen atom: Degeneracy

- States with different quantum numbers l and n are often referred to with letters as follows:

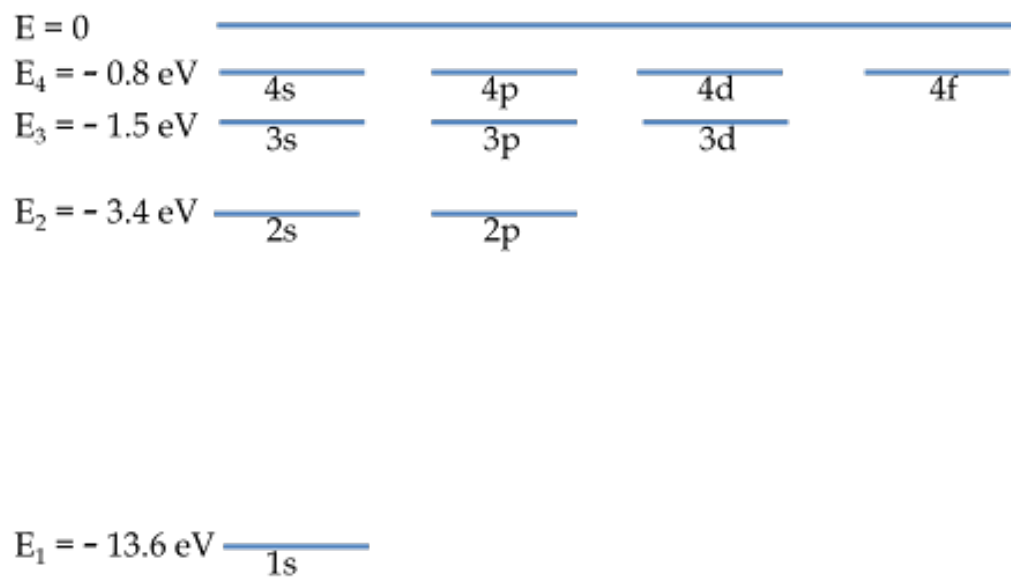
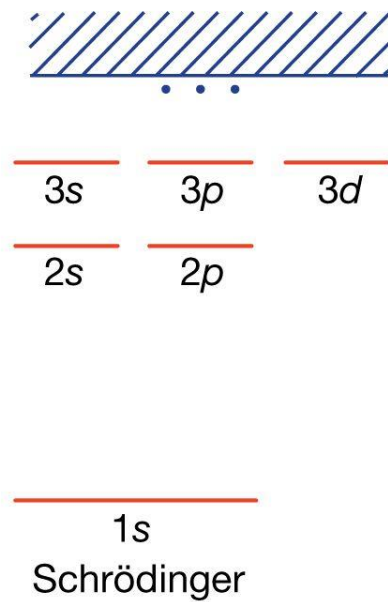
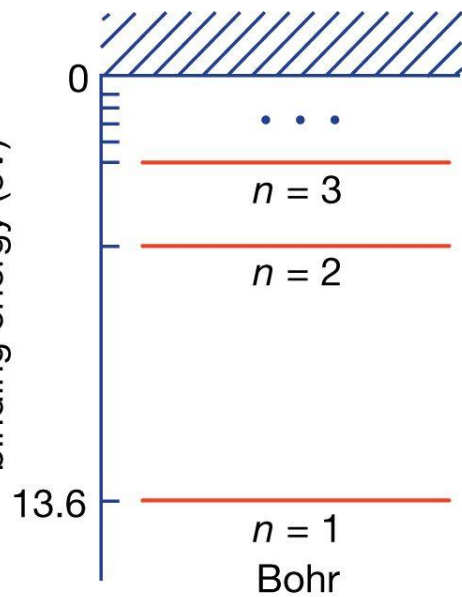
l value	letter
0	s
1	p
2	d
3	f
4	g
5	h

n value	shell
1	K
2	L
3	M
4	N

- Hydrogen atom states with **the same value of n but different values of l and m_l are degenerate** (have the same energy).
- Figures → radial probability distribution for states with $l = 0, 1, 2, 3$ and different values of $n = 1, 2, 3, 4$.



binding energy (eV)



RADIATIVE TRANSITIONS

Bohr postulated that the frequency of the radiation emitted by an atom dropping from an energy level E_m to a lower level E_n is

$$\nu = \frac{E_m - E_n}{h}$$

the time-dependent wave function Ψ_n of an electron in a state of quantum number n and energy E_n is the product of a time-independent wave function ψ_n and a time-varying function whose frequency is

$$\nu_n = \frac{E_n}{h}$$

$$\Psi_n = \psi_n e^{-\left(\frac{iE_n}{\hbar}\right)t}$$

$$\Psi_n^* = \psi_n^* e^{+\left(\frac{iE_n}{\hbar}\right)t}$$

The expectation value $\langle x \rangle$ of the position of such an electron is

$$\begin{aligned}\langle x \rangle &= \int_{-\infty}^{\infty} x \Psi_n^* \Psi_n dx = \int_{-\infty}^{\infty} x \psi_n^* \psi_n e^{[(iE_n/\hbar) - (iE_n/\hbar)]t} dx \\ &= \int_{-\infty}^{\infty} x \psi_n^* \psi_n dx\end{aligned}$$

The expectation value $\langle x \rangle \rightarrow$ constant in time since Ψ_n and Ψ_n^* are, by definition, functions of position only.

The electron does not oscillate, and no radiation occurs. Thus quantum mechanics predicts that a system in a specific quantum state does not radiate, as observed.

An electron that shifts from one energy state to another

The wave function Ψ_n of an electron that can exist in both states n and m

$$\Psi = a\Psi_n + b\Psi_m$$

a^*a = probability that the electron in state n

b^*b = probability that the electron in state m

$$a^*a + b^*b = 1$$

Initially $a=1$ and $b=0$;

when the electron is in the excited state $\rightarrow a=0$ and $b=1$; and ultimately $a=1$ and $b=0$ once more.

While the electron is in either state, there is no radiation, but when it is in the midst of the transition from m to n (that is, when both a and b have nonvanishing values), electromagnetic waves are produced.

The expectation value $\langle x \rangle$ that corresponds to the composite wave function

$$\begin{aligned}\langle x \rangle &= \int_{-\infty}^{\infty} x(a^* \Psi_n^* + b^* \Psi_m^*)(a \Psi_n + b \Psi_m) dx \\ &= \int_{-\infty}^{\infty} x(a^2 \Psi_n^* \Psi_n + b^* a \Psi_m^* \Psi_n + a^* b \Psi_n^* \Psi_m + b^2 \Psi_m^* \Psi_m) dx\end{aligned}$$

let $a^* a = a^2$ and $b^* b = b^2$.

The first and last integrals do not vary with time, so the second and third integrals are the only ones able to contribute to a time variation in $\langle x \rangle$.

$$\begin{aligned}\langle x \rangle = & a^2 \int_{-\infty}^{\infty} x \psi_n^* \psi_n dx + b^* a \int_{-\infty}^{\infty} x \psi_m^* e^{+(iE_m/\hbar)t} \psi_n e^{-(iE_n/\hbar)t} dx \\ & + a^* b \int_{-\infty}^{\infty} x \psi_n^* e^{+(iE_n/\hbar)t} \psi_m e^{-(iE_m/\hbar)t} dx + b^2 \int_{-\infty}^{\infty} x \psi_m^* \psi_m dx\end{aligned}$$

$$e^{i\theta} = \cos \theta + i \sin \theta \quad \text{and} \quad e^{-i\theta} = \cos \theta - i \sin \theta$$

$$\begin{aligned}\cos\left(\frac{E_m - E_n}{\hbar}\right)t \int_{-\infty}^{\infty} x [b^* a \psi_m^* \psi_n + a^* b \psi_n^* \psi_m] dx \\ + i \sin\left(\frac{E_m - E_n}{\hbar}\right)t \int_{-\infty}^{\infty} x [b^* a \psi_m^* \psi_n - a^* b \psi_n^* \psi_m] dx\end{aligned}$$

The real part of this result varies with time as

$$\cos\left(\frac{E_m - E_n}{\hbar}\right)t = \cos 2\pi\left(\frac{E_m - E_n}{h}\right)t = \cos 2\pi\nu t$$

Extra slide

Suppose the state is

$$\Psi(x, t) = a \psi_n(x) e^{-iE_n t/\hbar} + b \psi_m(x) e^{-iE_m t/\hbar}$$

where ψ_n, ψ_m are stationary-state eigenfunctions. If the state is normalized and the eigenstates are orthonormal,

$$|a|^2 + |b|^2 = 1.$$

Its complex conjugate is

$$\Psi^*(x, t) = a^* \psi_n^*(x) e^{+iE_n t/\hbar} + b^* \psi_m^*(x) e^{+iE_m t/\hbar}.$$

The expectation value of position is

$$\langle x \rangle = \int_{-\infty}^{\infty} \Psi^*(x, t) x \Psi(x, t) dx.$$

Extra slide

Substitute the two-state wavefunction:

$$\begin{aligned}\langle x \rangle = & \int \left[a^* \psi_n^* e^{+iE_n t/\hbar} + b^* \psi_m^* e^{+iE_m t/\hbar} \right] x \\ & \times \left[a \psi_n e^{-iE_n t/\hbar} + b \psi_m e^{-iE_m t/\hbar} \right] dx.\end{aligned}$$

Expanding gives four terms:

$$\begin{aligned}\langle x \rangle = & |a|^2 \int x |\psi_n|^2 dx + |b|^2 \int x |\psi_m|^2 dx \\ & + a^* b e^{-i(E_m - E_n)t/\hbar} \int \psi_n^* x \psi_m dx \\ & + b^* a e^{+i(E_m - E_n)t/\hbar} \int \psi_m^* x \psi_n dx.\end{aligned}$$

$$x_{nn} = \int \psi_n^* x \psi_n dx, \quad x_{mm} = \int \psi_m^* x \psi_m dx,$$

and

$$x_{nm} = \int \psi_n^* x \psi_m dx.$$

Extra slide

Because the position operator is Hermitian,

$$x_{mn} = x_{nm}^*.$$

Therefore

$$\langle x \rangle = |a|^2 x_{nn} + |b|^2 x_{mm} + a^* b x_{nm} e^{-i\omega_{mn}t} + b^* a x_{nm}^* e^{+i\omega_{mn}t}$$

where

$$\omega_{mn} = \frac{E_m - E_n}{\hbar}.$$

If a , b , and the relevant matrix element x_{nm} are real, and it simplifies to

$$\langle x \rangle(t) = a^2 x_{nn} + b^2 x_{mm} + 2ab x_{nm} \cos\left(\frac{E_m - E_n}{\hbar} t\right).$$

Extra slide

Now use

$$\hbar = \frac{h}{2\pi}.$$

Then

$$\frac{E_m - E_n}{\hbar} = 2\pi \frac{E_m - E_n}{h}.$$

So

$$\cos\left(\frac{E_m - E_n}{\hbar} t\right) = \cos\left[2\pi \left(\frac{E_m - E_n}{h}\right) t\right].$$

Defining

$$\boxed{\nu_{mn} = \frac{E_m - E_n}{h}}$$

gives

$$\boxed{\langle x \rangle(t) \propto \cos(2\pi\nu_{mn}t)}$$

for the time-dependent part.

This is precisely the **Bohr frequency condition**:

$$h\nu_{mn} = E_m - E_n$$

Extra slide

Now notice that the last two terms are complex conjugates of each other. If

$$C = a^* b x_{nm},$$

then

$$C e^{-i\omega t} + C^* e^{+i\omega t} = 2 \operatorname{Re}(C e^{-i\omega t}).$$

Write

$$C = |C| e^{i\delta}.$$

Then

$$C e^{-i\omega t} = |C| e^{i(\delta - \omega t)},$$

so

$$C e^{-i\omega t} + C^* e^{+i\omega t} = 2 |C| \cos(\omega t - \delta).$$

Hence the fully general result is

$$\langle x \rangle(t) = |a|^2 x_{nn} + |b|^2 x_{mm} + 2 |a^* b x_{nm}| \cos(\omega_{mn} t - \delta)$$

with

$$\omega_{mn} = \frac{E_m - E_n}{\hbar}.$$

The electron's position therefore oscillates sinusoidally at the frequency

$$\nu = \frac{E_m - E_n}{h}$$

- When the electron is in state n or state m the expectation value of the electron's position is constant.
- When the electron is undergoing a transition between these states, its position oscillates with the frequency .
- Such an electron, of course, is like an electric dipole and radiates electromagnetic waves of the same frequency .
- This result is the same as that postulated by Bohr and verified by experiment.

SELECTION RULES

The general condition necessary for an atom in an excited state to radiate is that the integral

$$\int_{-\infty}^{\infty} x \psi_n \psi_m^* dx \quad \text{not be zero}$$

Integral is finite $\rightarrow$ **allowed transitions**

Integral is zero $\rightarrow$ **forbidden transitions**

In the case of the hydrogen atom, three quantum numbers are needed to specify the initial and final states involved in a radiative transition. If the principal, orbital, and magnetic quantum numbers of the initial state are n', l', m'_l , respectively, and those of the final state are n, l, m_l , and u represents either the x, y , or z coordinate, the condition for an allowed transition is

Allowed transitions

$$\int_{-\infty}^{\infty} u \psi_{n,l,m_l} \psi_{n',l',m'_l}^* dV \neq 0$$

Since the wave functions Ψ_{n,l,m_l} for the hydrogen atom are known, the allowed transitions can be evaluated for $u = x$, $u = y$, and $u = z$ for all pairs of states differing in one or more quantum numbers.

When this is done, it is found that the only transitions between states of different n that can occur are those in which the orbital quantum number l changes by 1 or -1 and the magnetic quantum number m_l does not change or changes by 1 or -1 .

Selection Rules

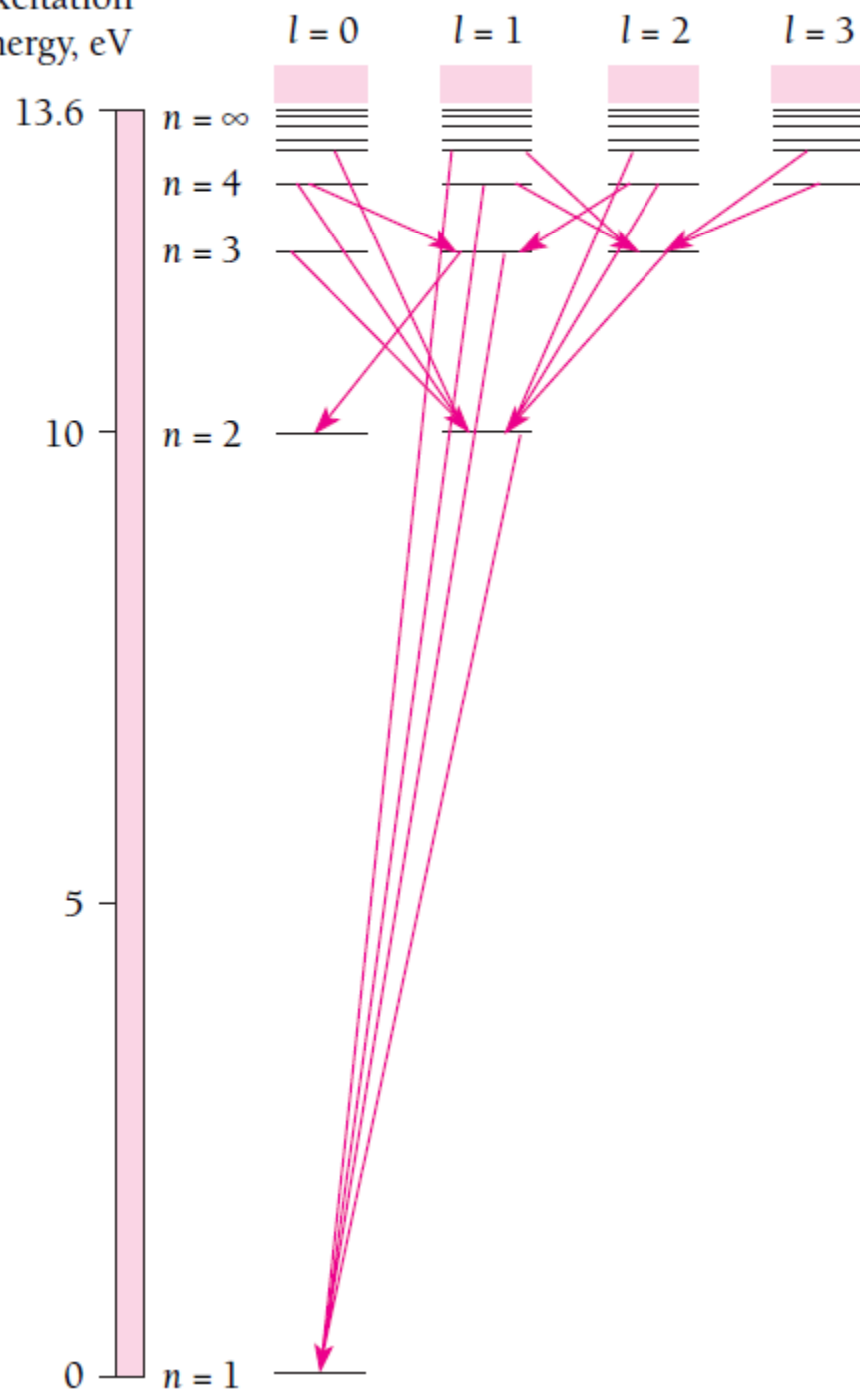
$$\Delta n = \text{not restricted}$$

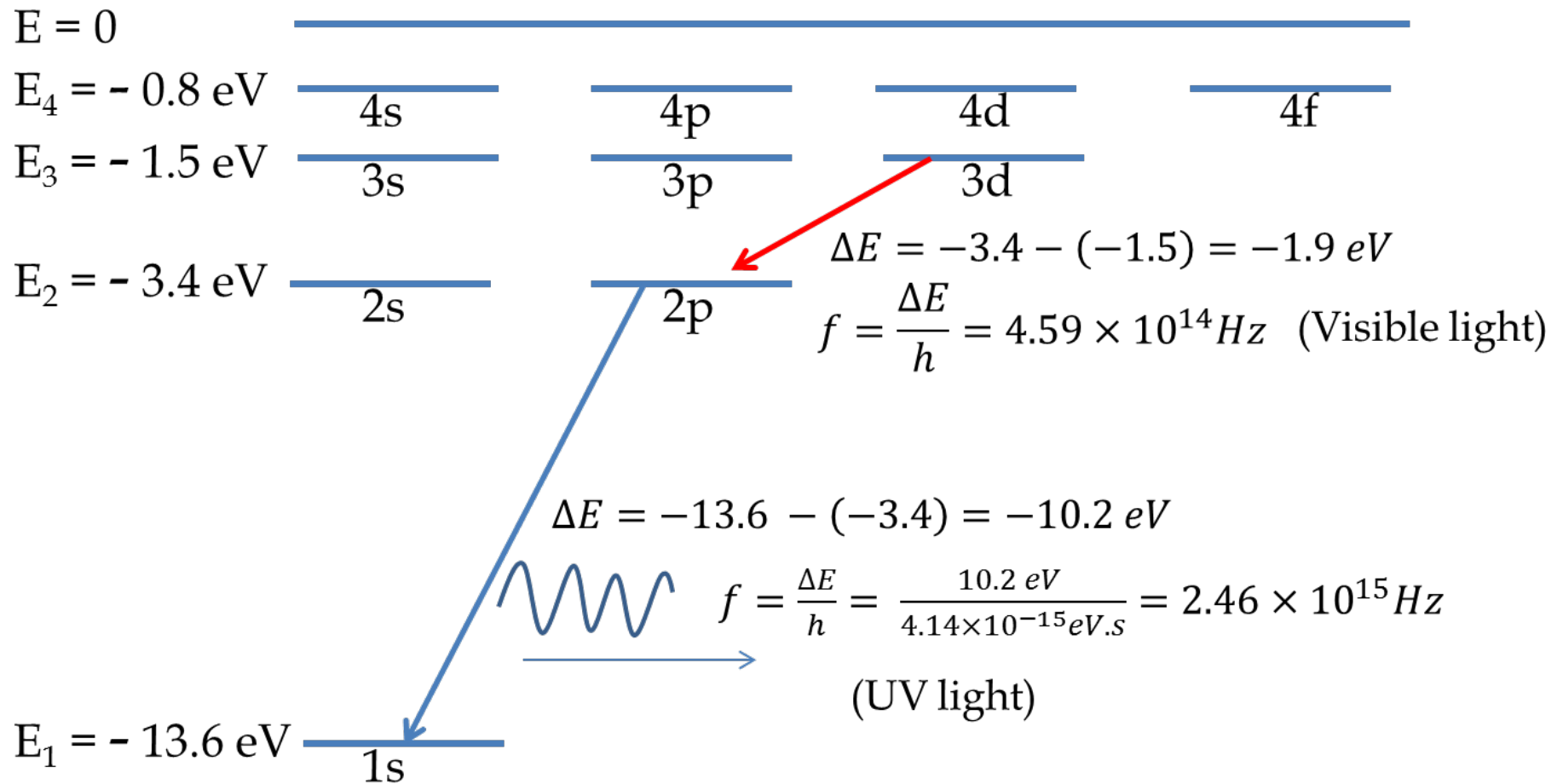
$$\Delta l = \pm 1 \text{ (For all polarization of light)}$$

$$\Delta m_l = 0 \text{ (For linearly polarization of light)}$$

$$\Delta m_l = \pm 1 \text{ (For circular polarization of light)}$$

Excitation
energy, eV

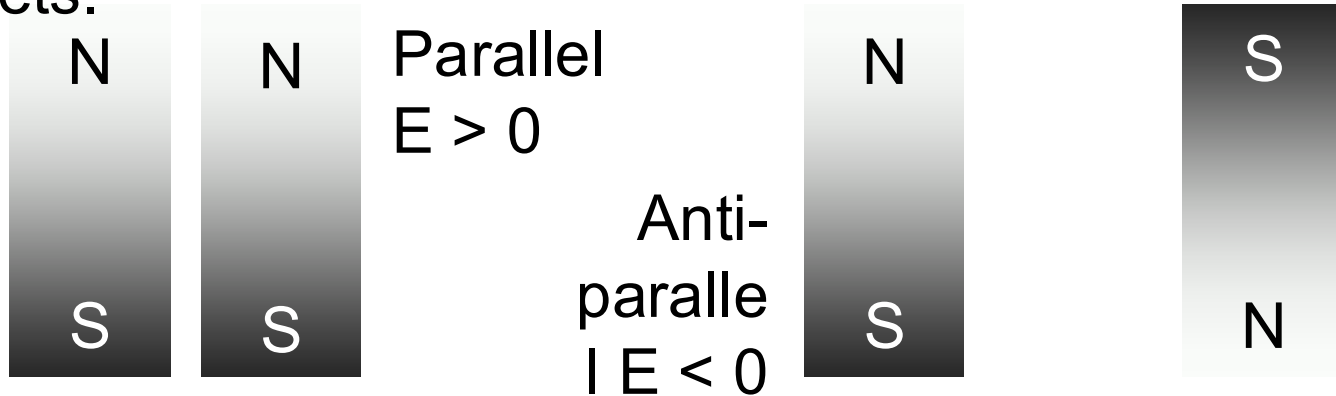




The Zeeman effect and selection rules

The Hydrogen Atom in a Magnetic Field

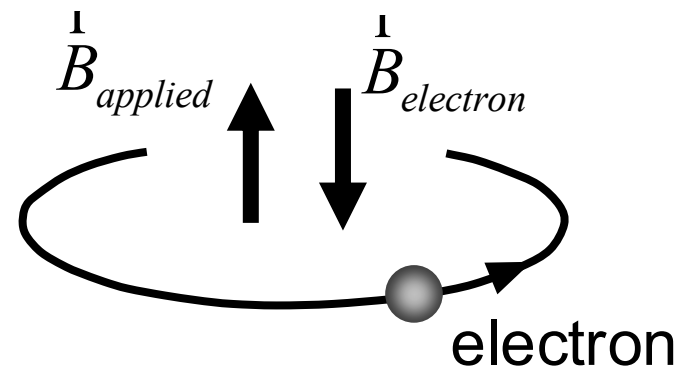
Consider pairs of magnets.



So, parallel magnets have more energy than anti-parallel magnets.

An electron orbiting around a nucleus generates a magnetic field.

So, an H atom and an applied magnetic field are two magnets.



Magnetic Effects: The Zeeman Effect

In 1896, the Dutch physicist Pieter Zeeman showed that spectral lines emitted by atoms in a **magnetic field** split into multiple energy levels.

Think of an electron as an orbiting circular current loop of $I = dq/dt$ around the nucleus.

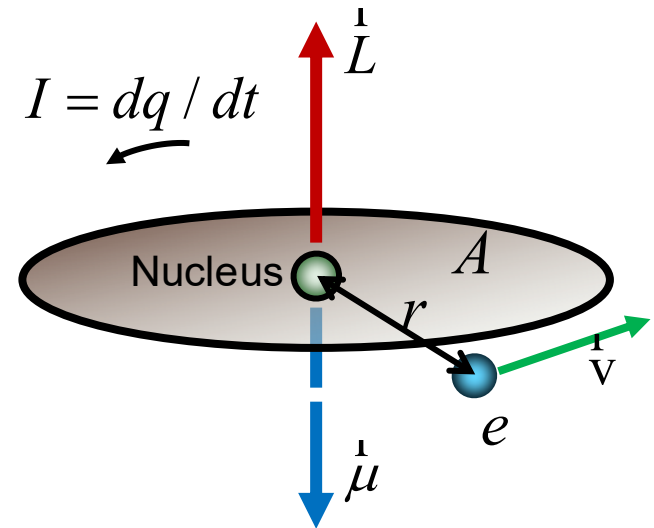
If the period is $T = 2\pi r/v$, then:

$$I = -e/T = -e/(2\pi r/v) = -ev/(2\pi r)$$

The current loop has a magnetic moment $\mu = IA$:

$$= [-ev/(2\pi r)]\pi r^2 = (-e/2m) mrv \quad \vec{\mu} = -\frac{e}{2m} \vec{L}$$

where $L = mvr$ is the magnitude of the orbital angular momentum.

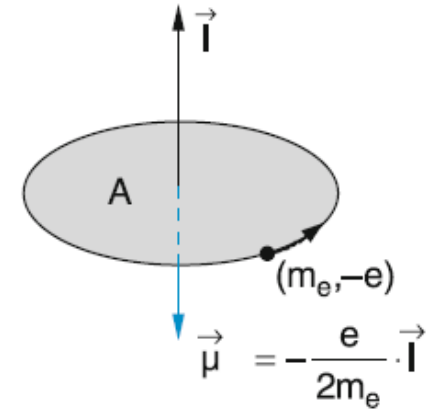


In an external magnetic field the potential energy of a magnetic dipole with magnetic moment μ is

$$E_{pot} = -\mu \cdot \mathbf{B}$$

$$= +\frac{e}{2m_e} \mathbf{l} \cdot \mathbf{B}$$

$$\mu = -\frac{e}{2m_e} \mathbf{l}$$

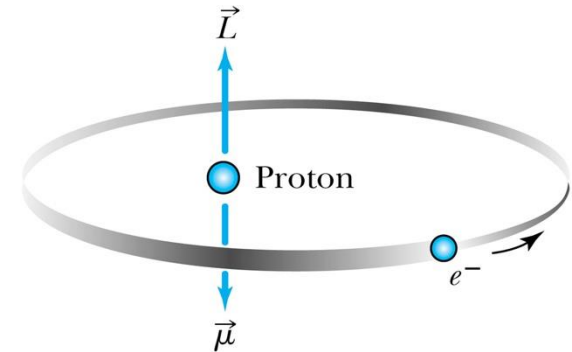


When the magnetic field points into the z -direction ($\mathbf{B} = \{0, 0, B_z = B\}$), because of $l_z = m\hbar$,

$$E_{pot} = +\frac{e\hbar}{2m_e} mB$$

where m (the projection of $\mathbf{l}$ onto the z -axis) is called the magnetic quantum number, It can take values $-l \leq m \leq +l$.

$$\mu_B = \frac{e\hbar}{2m_e} = 9.274015 \times 10^{-24} \text{ J/T} \rightarrow \text{Bohr magneton}$$



The additional energy caused by the magnetic field as

$$\Delta E_m = \mu_B m B$$

This gives for the energies of the hydrogen atomic states in an external magnetic field

$$E_{n,l,m} = E_{\text{Coul}}(n) + \mu_B m B$$

The $2l + 1$ *m*-sublevels that are degenerate without magnetic field split into $2l + 1$ **equidistant** Zeeman components with energies between

$$E_n^{min} = E_{coul}(n) - \mu_B mB$$

$$E_n^{max} = E_{coul}(n) + \mu_B mB$$

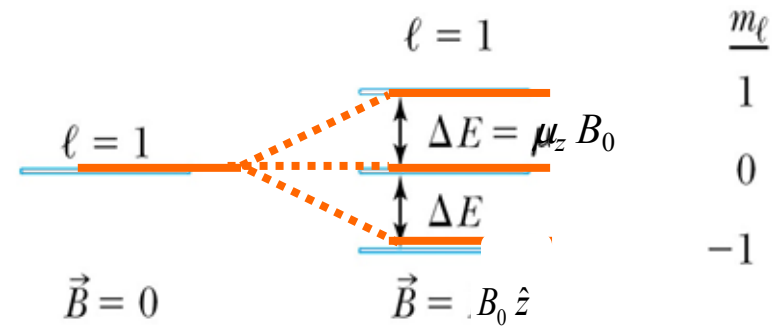
With an energetic distance

$$\Delta E = E_{n,l,m} - E_{n,l,m-1} = \mu_B B$$

between adjacent levels which is determined by the product of Bohr Magneton μ_B and magnetic field strength $\mathbf{B}$.

$$\mu_B = \frac{e\hbar}{2m_e}$$

The splitting of the $2l+1$ degenerate m components in an external magnetic field B due to the orbital magnetic moment related to the angular momentum $|l| = \sqrt{l(l+1)}\hbar$ is called **normal Zeeman effect**.



Zeeman splitting of a level with $l=1$ in a homogeneous magnetic field (normal Zeeman effect)

m_ℓ	Energy
1	$E_0 + \mu_B B$
0	E_0
-1	$E_0 - \mu_B B$

$$\mu_B = \frac{e\hbar}{2m_e}$$

$$E_1 = E_{coul}(n) - \mu_B B$$

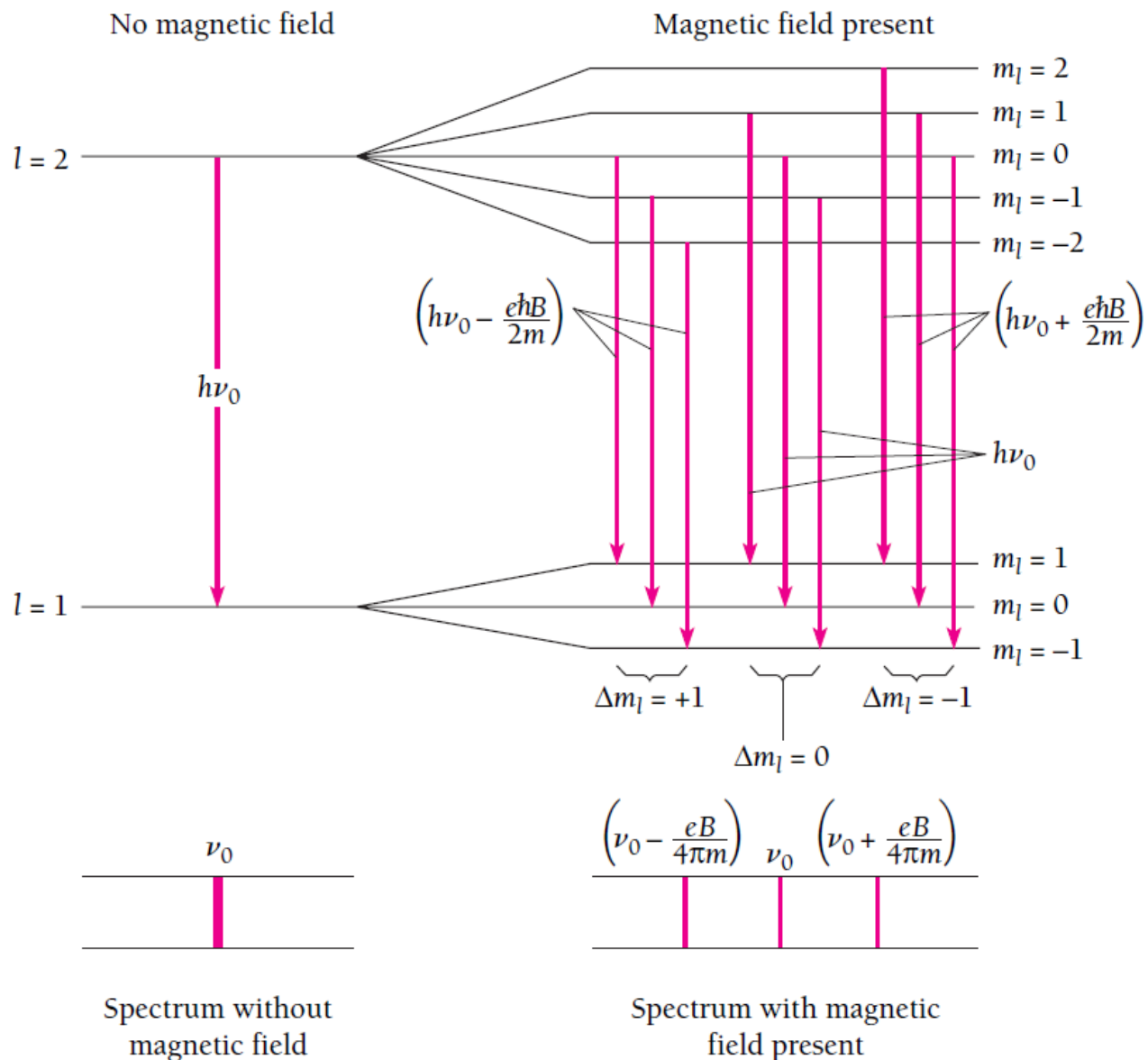
$$v_1 = v_{coul}(n) - \frac{\mu_B B}{h}$$

$$= v_{coul}(n) - \frac{e\hbar}{4\pi m_e} \frac{B}{h}$$

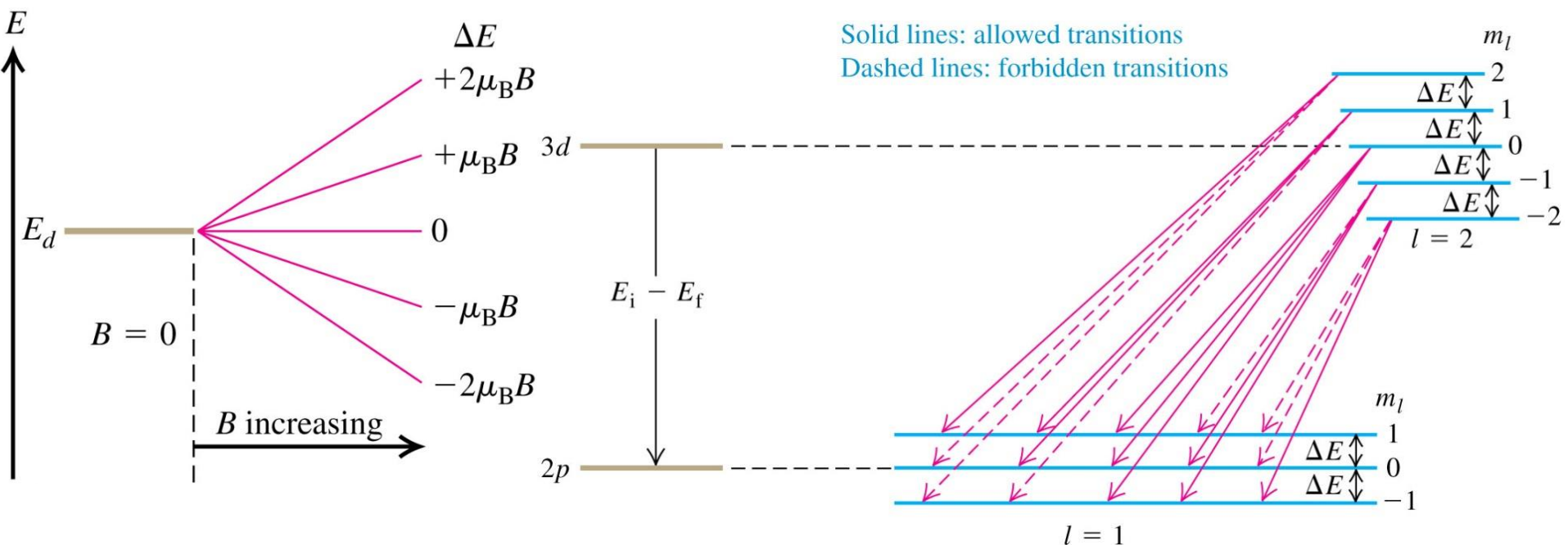
$$= v_{coul}(n) - \frac{eB}{4\pi m_e}$$

$$v_2 = v_{coul}(n)$$

$$v_3 = v_{coul}(n) + \frac{eB}{4\pi m_e}$$

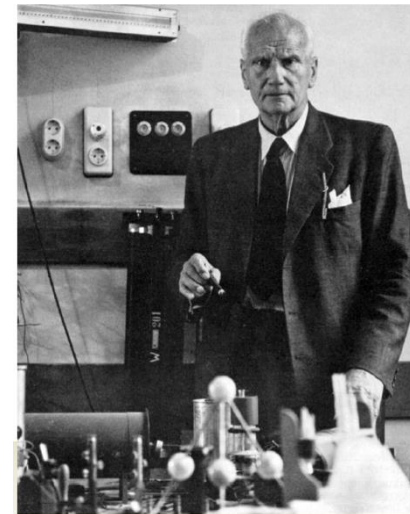
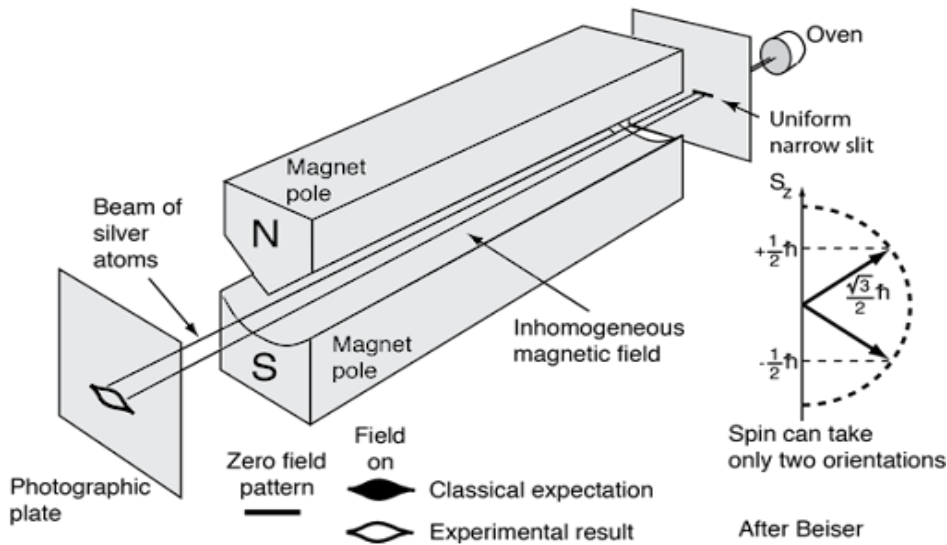
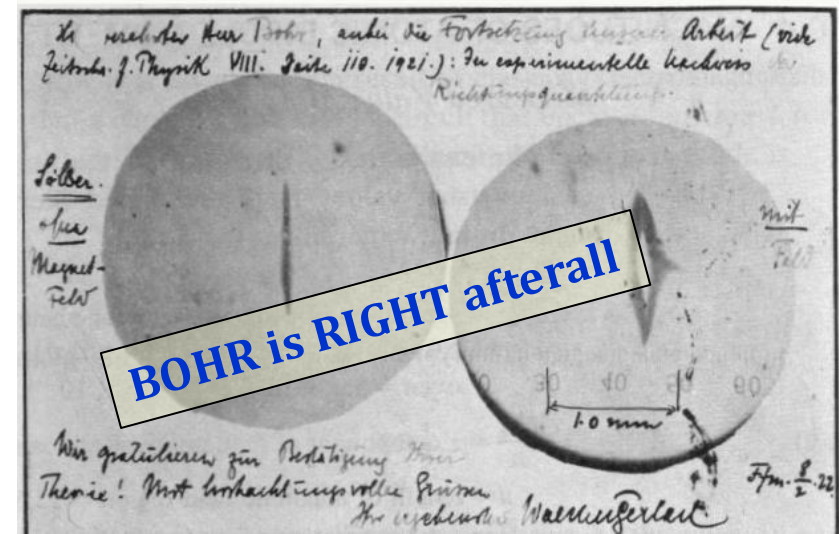
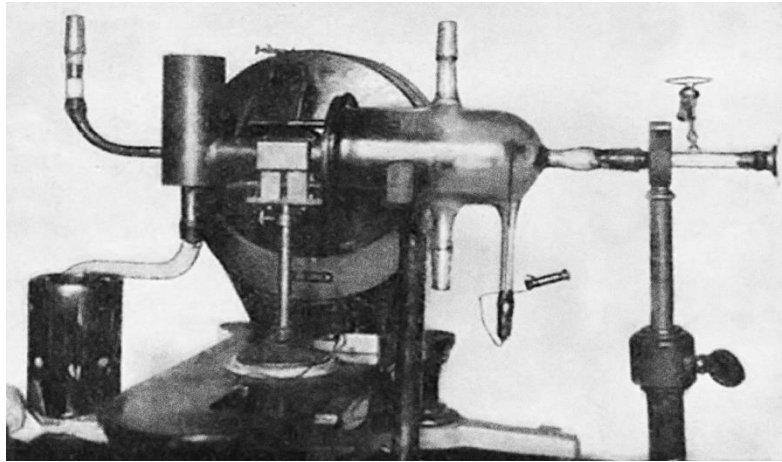


- Lower left figure shows the shift in energy of the five $l = 2$ states (each with a different value of m_l) as the magnetic field strength is increased.
- An atom in a magnetic field can make transitions between different states by emitting or absorbing a photon. A transition is *allowed* if l changes by 1 (one) and m_l changes by 0, 1, or -1 . (This is because a photon itself carries angular momentum.)
- A transition is *forbidden* if it violates these *selection rules*.



Stern-Gerlach (SG) experiment : Spin of electron

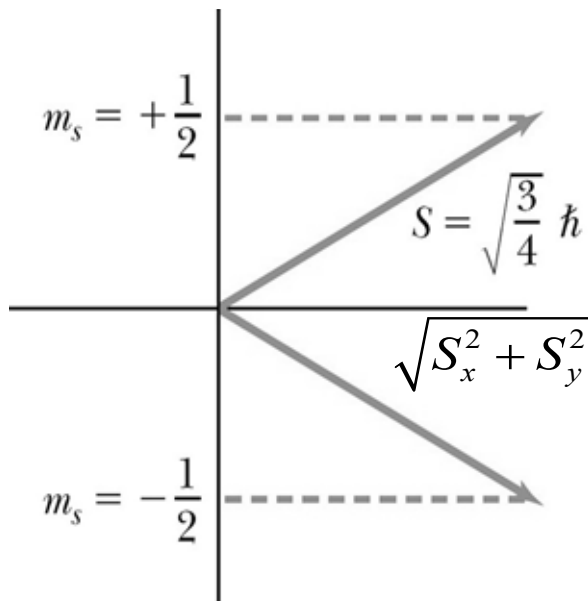
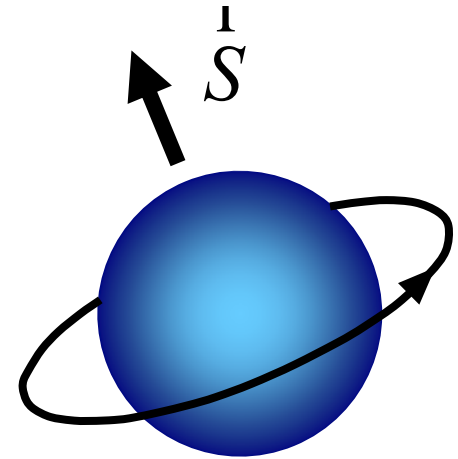
Stern-Gerlach experiment could be explained if assumes that electron has an intrinsic spin and spin must have a value $\frac{1}{2}$.



Intrinsic Spin of Electrons

In analogy with orbital angular momentum, the total spin angular momentum is:

$$|\vec{S}| = \sqrt{s(s+1)} \hbar = \sqrt{3/4} \hbar$$



The z -component of the spinning electron is also analogous to that of the orbiting electron:

So:

$$S_z = m_s \hbar$$

Because the **spin quantum number** m_s has only two values, $\pm\frac{1}{2}$, we say that the electron's spin is either “up” ($m_s = +\frac{1}{2}$) or “down” ($m_s = -\frac{1}{2}$).

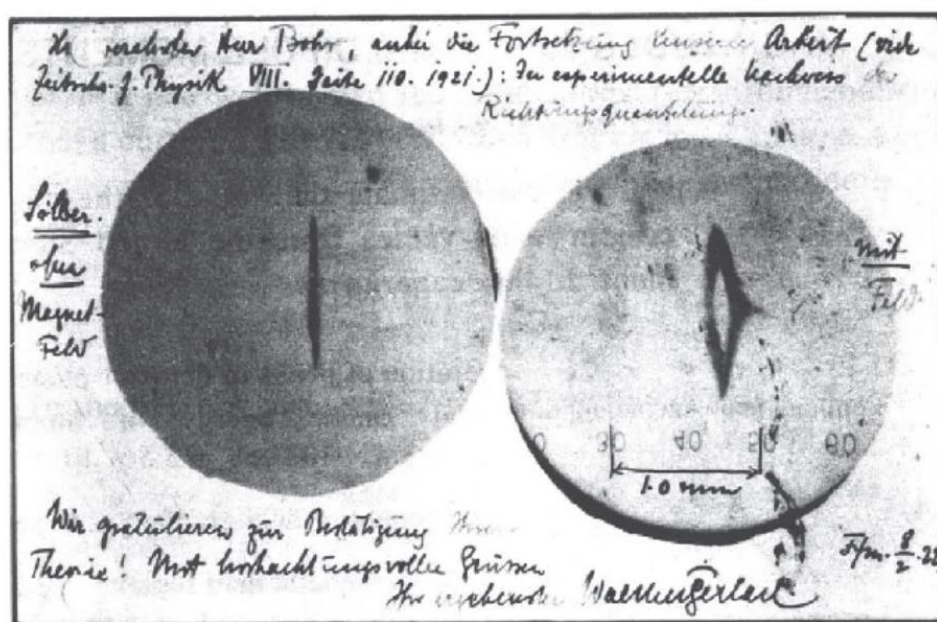


Figure 2: Postcard sent by Gerlach to Bohr

In case of doubt write to me:
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ZEEMAN EFFECT

H atom in an external magnetic field $\rightarrow$ semiclassical model
the electron motion is described by a classical circular orbit, while the angular momentum is given by the quantum mechanical expression

$$|l| = \sqrt{l(l+1)} \hbar$$

An electron $\rightarrow -e$, moving with the velocity $\rightarrow v$,
 the circular frequency $\rightarrow \nu = v/2\pi r$ on a circle $r \rightarrow$ radius
 The generated electric current

$$I = -ev = -\frac{ev}{2\pi r}$$

velocity

The current causes a magnetic moment

$$\boldsymbol{\mu} = I\mathbf{A} = I\pi r^2 \hat{n}$$

Where $\mathbf{A} = \pi r^2 \hat{n}$ the area vector
 perpendicular to the plane of the motion

The angular momentum of the circulating electron is

$$\mathbf{l} = \mathbf{r} \times \mathbf{p} = m_e r v \hat{n}$$

$$\boldsymbol{\mu} = -\frac{e}{2m_e} \mathbf{l}$$

Since μ is proportional to l , the orbital magnetic moment is often labeled μ_l .

